

employees.csv

emp_id	emp_name	department	job_title	salary	location	hire_date
101	Ramesh	IT	Data Engineer	95000	Bangalore	2019-06-15
102	Anita	HR	HR Executive	65000	Hyderabad	2020-02-10
103	John	Finance	Finance Manager	110000	New York	2016-09-01
104	Meena	IT	Senior Data Analyst	98000	Bangalore	2018-11-23
105	David	IT	Software Engineer	90000	San Francisco	2021-07-05
106	Priya	HR	HR Business Partner	72000	Hyderabad	2019-03-18
107	Michael	Finance	Account Director	120000	New York	2015-01-12
108	Sneha	IT	Junior Developer	78000	Bangalore	2022-08-09
109	Arjun	IT	Data Architect	115000	Bangalore	2017-05-30
110	Emily	Marketing	Marketing Manager	88000	Chicago	2018-10-17

departments.csv

dept_id	department	manager	budget	location
10	IT	Arjun Rao	500000	Bangalore
20	HR	Anita Sharma	200000	Hyderabad
30	Finance	John Smith	600000	New York
40	Marketing	Emily Wilson	300000	Chicago

Snowflake Cloning – Practical Assignments

Important Rule for Students:

-  No CTAS
 -  No data reload
 -  Only cloning & observation
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Assignment 1: Base Table Analysis

- Load both datasets into Snowflake tables
 - Validate row counts and data distribution
 - Identify candidate tables for cloning
 - Explain why these tables are suitable for cloning experiments
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Assignment 2: Table-Level Cloning

- Clone the employee table
 - Compare:
 - Row count
 - Query performance
 - Storage usage
 - Explain why clone creation is instantaneous
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Assignment 3: Change Isolation Testing

- Update salary data in the original employee table
 - Delete records in the cloned table
 - Insert new records in the clone
 - Validate isolation between original and clone
 - Explain backend copy-on-write behavior
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Assignment 4: Storage Growth Exploration

- Perform DML operations incrementally
 - Track storage before and after each operation
 - Identify the exact moment storage increases
 - Explain Snowflake's physical storage strategy
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Assignment 5: Time Travel Cloning

- Perform multiple updates on the original table
 - Create clones from:
 - Current state
 - Past state (Time Travel)
 - Compare data across versions
 - Explain real-world recovery use cases
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Assignment 6: Schema-Level Cloning

- Clone a schema containing employee and department tables

- Modify tables in cloned schema
 - Drop tables in original schema
 - Observe behavior and explain metadata inheritance
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Assignment 7: Database-Level Cloning

- Clone the entire database
 - Test query access and object availability
 - Perform DML on cloned database
 - Explain how Snowflake isolates environments (Dev/Test/Prod)
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Assignment 8: Privileges & Security

- Analyze access privileges before cloning
 - Observe privileges after cloning
 - Modify privileges on clone
 - Explain why Snowflake handles security this way
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Assignment 9: Performance Comparison

- Run identical analytical queries on:
 - Original tables
 - Cloned tables
 - Compare execution plans
 - Explain performance similarities/differences
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Assignment 10: Cleanup & Retention

- Drop original objects
- Validate cloned object availability
- Drop clone and analyze Time Travel behavior
- Explain lifecycle and retention rules